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--- Summary ---

NASIONALE
SENIOR SERTIFIKAAT

GRAAD 11

NOVEMBER 2013

WISKUNDIGE GELETTERDHEID V2
MEMORANDUM

PUNTE : 100

Simbool Verduideliking

M Metode

MA Metode met akkuraatheid

CA Voortgesette akkuraatheid

A Akkuraatheid (Antwoord)

C Omskakeling

S Vereenvoudiging

RT/RG/RM Lees van tabel/Lees van grafiek/ Lees van kaart

F Kies van korrekte formule

SF Vervanging in formule

J Mening

P Penalisering vir geen eenhede, verkeerde afronding, ens.

x ■

= 182,5

= 182 da e ■

1:M

1:M

1:A

1:M

1:M

1:A (3)

1.1.5 Bedrag per dag = 2 477 ,28 x 12 ■

=

■

= R81,44 ■ 1:M

1:M

1A (3)

1.1.6 WVF vir 6 maande = 182 da e x 81,44 ■

= R 14 822,08 ■ 1:M

1:A (2)

$$\begin{aligned} 1.1.7 \text{ Laaste betaling} &= 2\,477,28 \times 5 \\ &= R12\,386,40 \blacksquare \\ &= R14\,822,08 - R12\,386,40 \blacksquare \\ &= R2\,435,68 \blacksquare 1:MA \end{aligned}$$

1:M

1:A (3)

(NOVEMBER 2013) MATHEMATICAL LITERACY P2 3

$$\begin{aligned} 1.2 \text{ 1.2.1 Buite -Opp} &= 2 \times \text{Opp van Basis} + \text{Omtrek van Basis} \times \text{Hoogte} \\ &= 2 \times \blacksquare r^2 + \blacksquare d \times h \\ &= (2 \times 3,142 \times 4,752) + (3,142 \times 9,5 \text{ cm}) \times 17,8 \text{ cm} \blacksquare \blacksquare \\ &= 141,78275 \text{ cm}^2 + 29,849 \text{ cm} \times 17,8 \text{ cm} \\ &= 141,78275 \text{ cm}^2 + 531,3122 \text{ cm}^2 \blacksquare \\ &= 673,09495 \text{ cm}^2 \\ &= 673,09 \text{ cm}^2 \blacksquare 2:SF \end{aligned}$$

1:S

1:A (4)

$$\begin{aligned} 1.2.2 \text{ Buite -Opp} &= 2 \times \text{Opp van basis} + \text{Omtrek van Basis} \times \text{Hoogte} \\ &= 2 \times l \times b + 2 \times (l + b) \times h \\ &= (2 \times 18,7 \times 7,2) + 2(18,7 + 7,2) \times 3,8 \blacksquare \blacksquare \\ &= 269,28 \text{ cm}^2 + 51,8 \text{ cm} \times 3,8 \text{ cm} \\ &= 269,28 \text{ cm}^2 + 196,84 \text{ cm}^2 \blacksquare \\ &= 466,12 \text{ cm}^2 \blacksquare \end{aligned}$$

2:SF

1:S

1:A (4)

1.2.3 Opp.

$\blacksquare \blacksquare$ 1:A

2:R (3)

4 MATHEMATICAL LITERACY P2 (NOVEMBER 2013)

2.2 2.2.1 2008 $\blacksquare$ – 03:35:25
Drie uur vyf en dertig minute en vyf en twintig sekondes $\blacksquare$ 1:A

1:A (2)

2.2.2 2010 $\blacksquare$
03:42:19 $\blacksquare$ 1:A
1:A (2)

2.2.3 In 2005 was Elena se tyd stadiger as in 2004, maar vanaf 2006 tot 2008 het haar klaarmaaktye verbeter en van 2009 na 2010 het haar pas $\blacksquare$ afname in 2012 .
 $\blacksquare \blacksquare$ 2:A (2)

2.3.2 $P = 90/100 \blacksquare \blacksquare = 0,9 \blacksquare$ 1:Nom mer
1:Teller
1:Desimaal (3)
[28]

VRAAG 3

3.1 3.1.1 Vorige lesing = Huidige lesing – Verbruik

$$= 185\,523 \text{ k} - 30,3 \text{ k}$$

$$= 185\,492,7 \text{ k} \quad 1:\text{M}$$

1:A (2)

3.1.2 Verbruikskoste =

$$(6 \times 0) + (9 \times 7,90) + (10 \times 8,90) + (5,3 \times 9,90)$$

$$= 0 + 71,10 + 89,00 + 52,47$$

$$= \text{R}212,57 \quad 1:\text{M}$$

1:S

1:A (3)

3.1.3 Rioleringskoste = R103,32

$$\text{R}103,32 = (4 \times 0) + (3 \times 3,54) + (8 \times 8,52) + (2 \times 12,27)$$

$$= 0 + 10,62 + 68,16 + 24,54$$

$$= \text{R}103,32$$

$$\text{Volume} = 4 + 3 + 8 + 2$$

$$= 17 \text{ k} \quad 1:\text{M}$$

1:M

1:A (3)

3.1.4 BTW = $(212,57 + 103,32) \times 0,14$

$$= \text{R}315,89 \times 0,14$$

$$= \text{R}44,22 \quad 1:\text{CA}$$

1:CA (2)

3.1.5 Totale koste = $212,57 + 103,32 + 44,22$

$$= \text{R}360,11 \quad 1:\text{CA}$$

1:CA (2)

3.2 3.2.1 Gemiddelde =

$$20,8 \text{ k} + 44,2 \text{ k} + 22,5 \text{ k} + 21,6 \text{ k} + 15,4 \text{ k} + 22,7 \text{ k} +$$

$$20,4 \text{ k} + 25,1 \text{ k} + 23,6 \text{ k} + 24,9 \text{ k} +$$

$$21,5 \text{ k} + 30,3 \text{ k}$$

$$12$$

$$= 293/12$$

$$= 24,42 \text{ k} \quad 1:\text{M}$$

1:M

1:A (3)

3.2.2 15,4 k ; 20,4 k ; 20,8 k ; 21,5 k ; 21,6 k ; 22,5 k ;

22,7 k ; 23,6 k ; 24,9 k ; 25,1 k ; 30,3 k ; 44,2 k

Medi aan =

=

■

$$= 22,6$$

1:M

1:MA

1:A (3)

3.2.3 Met verwysing na die staat kan dit wees dat die huisgesin met vakansie was .

x ■

$$= 182,5$$

$$= 182 \text{ days } \blacksquare$$

1:M

1:M

1:A

1:M

1:M

1:A (3)

$$1.1.5 \text{ Amount per day} = 2\,477,28 \times 12 \blacksquare$$

$$=$$

■

$$= R81,44 \blacksquare 1:M$$

1:M

1A (3)

$$1.1.6 \text{ UIF for six months} = 182 \text{ days} \times 81,44 \blacksquare$$

$$= R14\,822,08 \blacksquare 1:M$$

1:A (2)

$$1.1.7 \text{ Last payment} = 2\,477,28 \times 5$$

$$= R12\,386,40 \blacksquare$$

$$= R14\,822,08 - R12\,386,40 \blacksquare$$

$$= R2\,435,68 \blacksquare 1:MA$$

1:M

1:A (3)

(NOVEMBER 2013) MATHEMATICAL LITERACY P2 3

1.2 1.2.1 Surface Area = 2 x Area of Base + Perimeter of Base
x Height

$$= 2 \times \blacksquare r^2 + \blacksquare d \times h$$

$$= (2 \times 3,142 \times 4,752) + (3,142 \times 9,5 \text{ cm}) \times 17,8 \text{ cm } \blacksquare \blacksquare$$

$$= 141,78275 \text{ cm}^2 + 29,849 \text{ cm} \times 17,8 \text{ cm}$$

$$= 141,78275 \text{ cm}^2 + 531,3122 \text{ cm}^2 \blacksquare$$

$$= 673,09495 \text{ cm}^2$$

$$= 673,09 \text{ cm}^2 \blacksquare 2:SF$$

1:S

1:A (4)

1.2.2 Surface Area = 2 x Area of Base + Perimeter of Base
x Height

$$= 2 \times l \times b + 2 \times (l+b) \times h$$

$$= (2 \times 18,7 \times 7,2) + 2(18,7 + 7,2) \times 3,8 \blacksquare \blacksquare$$

$$= 269,28 \text{ cm}^2 + 51,8 \text{ cm} \times 3,8 \text{ cm}$$

$$= 269,28 \text{ cm}^2 + 196,84 \text{ cm}^2 \blacksquare$$

$$= 466,12 \text{ cm}^2 \blacksquare$$

2:SF

1:S

1:A (4)

1.2.3 Area of roll = L x B
= 25 m x 150 cm ■
= 2 500 cm x 150 cm ■
= 375 000 cm² ■

Number of pencil cases =

■

= 557 ,1320329
= 557 pencil cases ■

No, the roll of material will not be enough .

Account Details as at 07/01/20 13

Account No : 555 444 333

Water (Period 05/12/2012 to 04/01/2013 – 31 days) Actual Reading

At 72 Nowhere Street, Mountain View / ERF 9876543

Previous Reading:

Current Reading:

185 523

Consumption:

30,3 k ■

Consumption Charge

Sewage (Period 05/12/2012 to 04/01/2013 – 31 days) Actual Reading

At 72 Nowhere Street, Mountain View / ERF 9876543

Disposal Charge for _____ k ■: R103,32

VAT at 14%

R _____

Latest account total due:

R

WATER SEWAGE

Domestic 1: Tariffs Disposal Charges

Volume Tariff per k ■ Volume Tariff per k ■

< 6 k ■ Free < 4 k ■ Free

6,1 – 15 k ■ R7,90 4,1 – 7 k ■ R3,54

15,1 – 25 k ■ R8,90 7,1 – 15 k ■ R8,52

> 25 k ■ R9,90 > 15 k ■ R12,27

3.1.1 Calculate the previous reading.

--- Full Combined Text ---

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WISKUNDIGE GELETTERDHEID V2
MEMORANDUM

PUNTE : 100

Simbool Verduideliking
M Metode

MA Metode met akkuraatheid
 CA Voortgesette akkuraatheid
 A Akkuraatheid (Antwoord)
 C Omskakeling
 S Vereenvoudiging
 RT/RG/RM Lees van tabel/Lees van grafiek/ Lees van kaart
 F Kies van korrekte formule
 SF Vervanging in formule
 J Mening
 P Penaliserings vir geen eenhede, verkeerde afronding, ens.
 R (Afronding/Rede)

Hierdie memorandum bestaan uit 6 bladsye .

2 MATHEMATICAL LITERACY P2 (NOVEMBER 2013)

VRAAG 1

$$\begin{aligned}
 1.1.1.1 \text{ WVF} &= 0,01 \times 6\,240 \blacksquare \\
 &= R62,40 \times 2 \blacksquare \\
 &= R124,80 \blacksquare
 \end{aligned}$$

OF

$$\begin{aligned}
 \text{WVF} &= 0,02 \blacksquare \times 6\,240 \blacksquare \\
 &= R124,80 \blacksquare 1:M
 \end{aligned}$$

1:M

1:A

2:M

1:A (3)

1.1.2

$\blacksquare$

$$\begin{aligned}
 &= 29,2 + 10,52572682 \blacksquare \\
 &= 39,72572682 \\
 &= 39,7 \% \blacksquare 1:SF
 \end{aligned}$$

1:S

1:A (3)

$$\begin{aligned}
 1.1.3 \text{ Bedrag van WVF} &= 0,397 \times 6\,240 \blacksquare \\
 &= R2\,477,28 \blacksquare 1:M
 \end{aligned}$$

1:A (2)

$$\begin{aligned}
 1.1.4 \text{ Aantal dae} &= 365 \times 3 \blacksquare \\
 &=
 \end{aligned}$$

$\blacksquare$

$$\begin{aligned}
 &= 182,5 \\
 &= 182 \text{ dae} \blacksquare
 \end{aligned}$$

OF

$$\begin{aligned}
 \text{Aantal dae} &= \\
 \blacksquare
 \end{aligned}$$

$$\begin{aligned}
 &\dots \times \blacksquare \\
 &= 182,5 \\
 &= 182 \text{ dae} \blacksquare
 \end{aligned}$$

1:M

1:M

1:A

1:M

1:M

1:A (3)

$$\begin{aligned} 1.1.5 \text{ Bedrag per dag} &= 2\,477,28 \times 12 \blacksquare \\ &= \\ \blacksquare &= R81,44 \blacksquare 1:M \end{aligned}$$

1:M

1A (3)

$$\begin{aligned} 1.1.6 \text{ WVF vir 6 maande} &= 182 \text{ dae} \times 81,44 \blacksquare \\ &= R\,14\,822,08 \blacksquare 1:M \end{aligned}$$

1:A (2)

$$\begin{aligned} 1.1.7 \text{ Laaste betaling} &= 2\,477,28 \times 5 \\ &= R12\,386,40 \blacksquare \\ &= R14\,822,08 - R12\,386,40 \blacksquare \\ &= R2\,435,68 \blacksquare 1:MA \end{aligned}$$

1:M

1:A (3)

(NOVEMBER 2013) MATHEMATICAL LITERACY P2 3

$$\begin{aligned} 1.2 \text{ 1.2.1 Buite -Opp} &= 2 \times \text{Opp van Basis} + \text{Omtrek van} \\ &\quad \text{Basis} \times \text{Hoogte} \\ &= 2 \times \blacksquare r^2 + \blacksquare d \times h \\ &= (2 \times 3,142 \times 4,752) + (3,142 \times 9,5 \text{ cm}) \times 17,8 \text{ cm} \blacksquare\blacksquare \\ &= 141,78275 \text{ cm}^2 + 29,849 \text{ cm} \times 17,8 \text{ cm} \\ &= 141,78275 \text{ cm}^2 + 531,3122 \text{ cm}^2 \blacksquare \\ &= 673,09495 \text{ cm}^2 \\ &= 673,09 \text{ cm}^2 \blacksquare 2:SF \end{aligned}$$

1:S

1:A (4)

$$\begin{aligned} 1.2.2 \text{ Buite -Opp} &= 2 \times \text{Opp van basis} + \text{Omtrek van Basis} \times \\ &\quad \text{Hoogte} \\ &= 2 \times l \times b + 2 \times (l + b) \times h \\ &= (2 \times 18,7 \times 7,2) + 2(18,7 + 7,2) \times 3,8 \blacksquare\blacksquare \\ &= 269,28 \text{ cm}^2 + 51,8 \text{ cm} \times 3,8 \text{ cm} \\ &= 269,28 \text{ cm}^2 + 196,84 \text{ cm}^2 \blacksquare \\ &= 466,12 \text{ cm}^2 \blacksquare \end{aligned}$$

2:SF

1:S

1:A (4)

$$\begin{aligned} 1.2.3 \text{ Opp. van rol} &= l \times b \\ &= 25 \text{ m} \times 150 \text{ cm} \blacksquare \\ &= 2\,500 \text{ cm} \times 150 \text{ cm} \blacksquare \\ &= 375\,000 \text{ cm}^2 \blacksquare \end{aligned}$$

$$\begin{aligned} \text{Aantal potloodsakkies} &= \\ \blacksquare & \end{aligned}$$

$$= 557,1320329$$

= 557 potloodsakkies ■

Nee, die rol materiaal sal nie genoeg wees nie . ■ 1:SF

1:C

1:A

1:M

1:A

1:O (6)

[33]

VRAAG 2

2.1 2.1.1 21,1 km ■ 1:A (1)

2.1.2 49 meters ■■ 2:A (2)

2.1.3 Hoogste = 200 m

Laagste = 36 m ■

Verskil = 200 m – 36 m

= 164 m ■ 1:RG

1:A (2)

2.1.4 Tussen ±2 en ±4 km, ■ omdat die hoogte toeneem
vanaf ±38 m tot ± 140 m. (steil gradiënt .) ■■

OF

Tussen ±9 en ±13 km, ■ omdat die hoogte toeneem
vanaf ±60 m tot ±195 m (steil gradiënt). ■■ 1:A

2:R (3)

4 MATHEMATICAL LITERACY P2 (NOVEMBER 2013)

2.2 2.2.1 2008 ■ – 03:35:25

Drie uur vyf en dertig minute en vyf en twintig
sekondes ■ 1:A

1:A (2)

2.2.2 2010 ■

03:42:19 ■ 1:A

1:A (2)

2.2.3 In 2005 was Elena se tyd stadiger as in 2004,
maar vanaf 2006 tot 2008 het haar klaarmaaktye
verbeter en van 2009 na 2010 het haar pas

fg t ■ v b t g ■

afname in 2012 . ■■

(Aanvaar enige logiese verduideliking .) 2:A (2)

2.2.4 Totale Tyd =

03:37:51 + 03:38:12 + 03:36:29 + 03:35:34 +

03:35:25 + 03:40:43 + 03:42:19 + 03:37:54 +

03:41:56 ■

= 27h 341min 323sek ■

= 32h 46min 23sek ■■ 1:M

1:A

2:A (4)

2.2.5 03:42:19 ; 03:41:56 ; 03:40:43 ; 03:38:12 ;
 03:37:54 ; 03:37:51 ; 03:36:29 ; 03:35:34 ;
 03:35:25 ■
 Mediaan = 03:37:54 ■ 1:M
 1:A (2)

2.2.6 Verskil = 03:42:19 – 03:35:25 ■
 = 6min 54 sek ■■ 1:M
 2:A (3)

2.3 2.3.1 Dit beteken dat sy 90 uit 100 kanse het om die
 marathon te wen. ■■ 2:A (2)

2.3.2 $P = 90/100$ ■■ = 0,9 ■ 1:Nom mer
 1:Teller
 1:Desimaal (3)
 [28]

(NOVEMBER 2013) MATHEMATICAL LITERACY P2 5

VRAAG 3

3.1 3.1.1 Vorige lesing = Huidige lesing – Verbruik
 = 185 523 k ■ – 30,3 k ■ ■
 = 185 492,7 k ■ ■ 1:M
 1:A (2)

3.1.2 Verbruikskoste =
 $(6 \times 0) + (9 \times 7,90) + (10 \times 8,90) + (5,3 \times 9,90)$ ■
 = 0 + 71,10 + 89,00 + 52,47 ■
 = R212,57 ■ 1:M
 1:S
 1:A (3)

3.1.3 Rioleringskoste = R103,32
 $R103,32 = (4 \times 0) + (3 \times 3,54) + (8 \times 8,52) + (2 \times 12,27)$
 = 0 + 10,62 + 68,16 + 24,54 ■
 = R103,32
 Volume = 4 + 3 + 8 + 2 ■
 = 17 k ■ ■ 1:M

1:M
 1:A (3)

3.1.4 BTW = $(212,57 + 103,32) \times 0,14$
 = R315,89 x 0,14 ■
 = R44,22 ■ 1:CA
 1:CA (2)

3.1.5 Totale koste = 212,57 + 103,32 + 44,22 ■
 = R360,11 ■ 1:CA
 1:CA (2)

3.2 3.2.1 Gemiddelde =
 $20,8 \text{ k} \blacksquare + 44,2 \text{ k} \blacksquare + 22,5 \text{ k} \blacksquare + 21,6 \text{ k} \blacksquare + 15,4 \text{ k} \blacksquare + 22,7 \text{ k} \blacksquare +$
 $20,4 \text{ k} \blacksquare + 25,1 \text{ k} \blacksquare + 23,6 \text{ k} \blacksquare + 24,9 \text{ k} \blacksquare +$
 $21,5 \text{ k} \blacksquare + 30,3 \text{ k} \blacksquare \blacksquare$
 12 ■
 = 293/12
 = 24,42 k ■ ■ 1:M
 1:M

1:A (3)

3.2.2 15,4 k ■ ; 20,4 k ■ ; 20,8 k ■ ; 21,5 k ■ ; 21,6 k ■ ; 22,5 k ■ ;
22,7 k ■ ; 23,6 k ■ ; 24,9 k ■ ; 25,1 k ■ ; 30,3 k ■ ; 44,2 k ■
Medi aan =

$$\frac{15,4 + 20,4 + 20,8 + 21,5 + 21,6 + 22,5 + 22,7 + 23,6 + 24,9 + 25,1 + 30,3 + 44,2}{12} = 22,6$$

1:M

1:MA

1:A (3)

3.2.3 Met verwysing na die staat kan dit wees dat die huisgesin
met vakansie was . ■■
(Aanvaar enige relevante antwoord .) 2:O (2)

3.2.4 Lekkende kraan, ondergrondse lekkende pype, foutiewe
watermeter ■■
(Aanvaar enige relevante antwoord .) 2:O (2)

3.2.5 Mediaan , ■ omdat dit nie skeef getrek word deur die
uitskieters van 15,4 k ■ en 44,2 k ■ ■soos in die geval van
die gemiddelde . 2:O (2)
[24]

6 MATHEMATICAL LITERACY P2 (NOVEMBER 2013)

VRAAG 4

4.1 4.1.1 Die dollar is sterker . ■ Jy moet R9,04 vir elke \$1
betaal . ■ 1:A
1:R (2)

$$4.1.2 \text{ Rand waarde} = 100 \times 9,04 \text{ ■} \\ = \text{R}904 \text{ ■ 1:M}$$

1A (2)

4.1.3 Bankkoste word belas wanneer jy geld wissel van een
g h ■ . ■■ 2:A (2)

$$4.1.4 \text{ Persentasie} = \frac{904 - 858,80}{858,80} \times 100 \\ = 5\% \text{ ■ 1:M}$$

1:M

1:A (3)

4.2 Stap 1

Met die duime stoot die dekplaat uit om die battery te
verwyder . ■■

Stap 2

Haal die battery uit . ■■

Stap 3

Met die battery kompartement oop en die battery verwyder, sit die
Sim-kaart in die gleuf met die goue plaatjie na onder . ■■

(Aanvaar enige relevante verduidelikings .)

2:A

2:A

2:A

(6)
[15]

TOTAAL: 100

NATIONAL
SENIOR CERTIFICATE

GRADE 11

NOVEMBER 2013

MATHEMATICAL LITERACY P 2
MEMORANDUM

MARKS: 100

Symbol Explanation

M Method

MA Method with accuracy

CA Consistent accuracy

A Accuracy

C Conversion

S Simplification

RT/RG/RM Reading from a table/Reading from a graph/Read from map

F Choosing the correct formula

SF Substitution in a formula

J Justification

P Penalty, e.g. for no units, incorrect rounding off etc.

R Rounding Off/Reason

This memorandum consists of 6 pages .

2 MATHEMATICAL LITERACY P2 (NOVEMBER 2013)

QUESTION 1

$$\begin{aligned}1.1.1.1 \text{ UIF} &= 0,01 \times 6\,240 \blacksquare \\ &= R62,40 \times 2 \blacksquare \\ &= R124,80 \blacksquare\end{aligned}$$

OR

$$\begin{aligned}\text{UIF} &= 0,02 \blacksquare \times 6\,240 \blacksquare \\ &= R124,80 \blacksquare \quad 1:M\end{aligned}$$

1:M

1:A

2:M

1:A (3)

1.1.2

$$\begin{aligned}
 &= 29,2 + 10,52572682 \\
 &= 39,72572682 \\
 &= 39,7 \% \quad 1:SF
 \end{aligned}$$

1:S

1:A (3)

$$\begin{aligned}
 1.1.3 \text{ Amount from UIF} &= 0,397 \times 6\,240 \\
 &= R2\,477,28 \quad 1:M
 \end{aligned}$$

1:A (2)

$$\begin{aligned}
 1.1.4 \text{ Number of days} &= 365 \times 3 \\
 &= \\
 &= 182,5 \\
 &= 182 \text{ days}
 \end{aligned}$$

OR

Number of days =

$$\begin{aligned}
 &= 182,5 \\
 &= 182 \text{ days}
 \end{aligned}$$

1:M

1:M

1:A

1:M

1:M

1:A (3)

$$\begin{aligned}
 1.1.5 \text{ Amount per day} &= 2\,477,28 \times 12 \\
 &= \\
 &= R81,44 \quad 1:M
 \end{aligned}$$

1:M

1A (3)

$$\begin{aligned}
 1.1.6 \text{ UIF for six months} &= 182 \text{ days} \times 81,44 \\
 &= R14\,822,08 \quad 1:M
 \end{aligned}$$

1:A (2)

$$\begin{aligned}
 1.1.7 \text{ Last payment} &= 2\,477,28 \times 5 \\
 &= R12\,386,40 \\
 &= R14\,822,08 - R12\,386,40 \\
 &= R2\,435,68 \quad 1:MA
 \end{aligned}$$

1:M

1:A (3)

(NOVEMBER 2013) MATHEMATICAL LITERACY P2 3

1.2 1.2.1 Surface Area = $2 \times \text{Area of Base} + \text{Perimeter of Base} \times \text{Height}$

$$\begin{aligned}
&= 2 \times \pi r^2 + \pi d \times h \\
&= (2 \times 3,142 \times 4,752) + (3,142 \times 9,5 \text{ cm}) \times 17,8 \text{ cm} \\
&= 141,78275 \text{ cm}^2 + 29,849 \text{ cm} \times 17,8 \text{ cm} \\
&= 141,78275 \text{ cm}^2 + 531,3122 \text{ cm}^2 \\
&= 673,09495 \text{ cm}^2 \\
&= 673,09 \text{ cm}^2 \quad \text{2:SF}
\end{aligned}$$

1:S

1:A (4)

1.2.2 Surface Area = 2 x Area of Base + Perimeter of Base
x Height

$$\begin{aligned}
&= 2 \times l \times b + 2 \times (l+b) \times h \\
&= (2 \times 18,7 \times 7,2) + 2(18,7 + 7,2) \times 3,8 \\
&= 269,28 \text{ cm}^2 + 51,8 \text{ cm} \times 3,8 \text{ cm} \\
&= 269,28 \text{ cm}^2 + 196,84 \text{ cm}^2 \\
&= 466,12 \text{ cm}^2
\end{aligned}$$

2:SF

1:S

1:A (4)

$$\begin{aligned}
1.2.3 \text{ Area of roll} &= L \times B \\
&= 25 \text{ m} \times 150 \text{ cm} \\
&= 2 \text{ 500 cm} \times 150 \text{ cm} \\
&= 375 \text{ 000 cm}^2
\end{aligned}$$

Number of pencil cases =

$$\begin{aligned}
&= 557,1320329 \\
&= 557 \text{ pencil cases}
\end{aligned}$$

No, the roll of material will not be enough .

1:SF

1:C

1:A

1:M

1:A

1:O (6)

[33]

QUESTION 2

2.1 2.1.1 21,1 km 2:A (2)

2.1.2 49 meters 1:A (1)

2.1.3 Highest = 200 m

Lowest = 36 m

$$\begin{aligned}
\text{Difference} &= 200 \text{ m} - 36 \text{ m} \\
&= 164 \text{ m} \quad \text{1:RG}
\end{aligned}$$

1:A (2)

2.1.4 Between ± 2 and ± 4 km, because the height increases from ± 38 m to ± 140 m (steep gradient) .

OR

Between ± 9 and ± 13 km, because the height increases from ± 60 m to ± 195 m (steep gradient) .

2:R (3)

4 MATHEMATICAL LITERACY P2 (NOVEMBER 2013)

2.2.2.1 2008 ■ – 03:35:25

Three hours thirty five minutes and twenty five seconds ■ 1:A

1:A (2)

2.2.2 2010 ■

03:42:19 ■ 1:A

1:A (2)

2.2.3 In 2008 she was slower than in 2004, but from 2006 to 2008 she improved on her finishing times and then from 2009 to 2010 her pace slowed down with an improvement in 2011 and then slower in 2012. ■■

(Accept any logical explanation.) 2:A (2)

2.2.4 Total Time =

03:37:51 + 03:38:12 + 03:36:29 + 03:35:34 +
03:35:25 + 03:40:43 + 03:42:19 + 03:37:54 +
03:41:56 ■

= 27h 341min 323sec ■

= 32h 46min 23sec ■■ 1:M

1:A

2:A (4)

2.2.5 03:42:19 ; 03:41:56 ; 03:40:43 ; 03:38:12 ;
03:37:54 ; 03:37:51 ; 03:36:29 ; 03:35:34 ;
03:35:25 ■

Median = 03:37:54 ■ 1:M

1:A (2)

2.2.6 Difference = 03:42:19 – 03:35:25 ■

= 6min 54 sec ■■ 1:M

2:A (3)

2.3 2.3.1 It means she has 90 out of 100 chances to win the marathon. ■■ 2:A (2)

2.3.2 $P = \frac{90}{100} = 0,9$ ■ 1:Numerator

1:Denominator

1:Decimal (3)

[28]

(NOVEMBER 2013) MATHEMATICAL LITERACY P2 5

QUESTION 3

3.1 3.1.1 Previous reading = Current reading – consumption

= 185 523 k ■ – 30,3 k ■■

= 185 492,7 k ■■ 1:M

1:A (2)

3.1.2 Consumption charge

= $(6 \times 0) + (9 \times 7,90) + (10 \times 8,90) + (5,3 \times 9,90)$ ■

= 0 + 71,10 + 89,00 + 52,47 ■

= R212,57 ■ 1:M

1:S

1:A (3)

3.1.3 Disposal charge = R103,32

$$\begin{aligned}
 R103,32 &= (4 \times 0) + (3 \times 3,54) + (8 \times 8,52) + (2 \times 12,27) \\
 &= 0 + 10,62 + 68,16 + 24,54 \blacksquare \\
 &= R103,32 \\
 \text{Volume} &= 4 + 3 + 8 + 2 \blacksquare \\
 &= 17 \text{ k} \blacksquare \blacksquare 1:\text{M}
 \end{aligned}$$

1:M
1:A (3)

$$\begin{aligned}
 3.1.4 \text{ VAT} &= (212,57 + 103,32) \times 0,14 \\
 &= R 315,89 \times 0,14 \blacksquare \\
 &= R 44,22 \blacksquare 1:\text{CA} \\
 1:\text{CA} &(2)
 \end{aligned}$$

$$\begin{aligned}
 3.1.5 \text{ Total charge} &= 212,57 + 103,32 + 44,22 \blacksquare \\
 &= R 360,11 \blacksquare 1:\text{CA} \\
 1:\text{CA} &(2)
 \end{aligned}$$

$$\begin{aligned}
 3.2 \text{ 3.2.1 Mean} &= \\
 20,8 \text{ k} \blacksquare + 44,2 \text{ k} \blacksquare + 22,5 \text{ k} \blacksquare + 21,6 \text{ k} \blacksquare + 15,4 \text{ k} \blacksquare + 22,7 \text{ k} \blacksquare \\
 + 20,4 \text{ k} \blacksquare + 25,1 \text{ k} \blacksquare + 23,6 \text{ k} \blacksquare + 24,9 \text{ k} \blacksquare + \\
 21,5 \text{ k} \blacksquare + 30,3 \text{ k} \blacksquare \blacksquare \\
 12 \blacksquare \\
 &= 293/12 \\
 &= 24,42 \text{ k} \blacksquare \blacksquare 1:\text{M} \\
 1:\text{M}
 \end{aligned}$$

1:A (3)
3.2.2 15,4 k $\blacksquare$; 20,4 k $\blacksquare$; 20,8 k $\blacksquare$; 21,5 k $\blacksquare$; 21,6 k $\blacksquare$; 22,5 k $\blacksquare$; 22,7 k $\blacksquare$; 23,6 k $\blacksquare$; 24,9 k $\blacksquare$; 25,1 k $\blacksquare$; 30,3 k $\blacksquare$; 44,2 k $\blacksquare$ $\blacksquare$

Median =

$$\begin{aligned}
 &= \\
 \blacksquare &= 22,6 \blacksquare \\
 1:\text{M}
 \end{aligned}$$

1:MA

1:A (3)
3.2.3 Referring to the statement date, it can be that the residents were on vacation during this time . $\blacksquare \blacksquare$
(Accept any relevant answer .) 2:O (2)
3.2.4 Leaking tap, underground leaking pipes, faulty water meter $\blacksquare \blacksquare$
Accept any relevant answer 2:O (2)
3.2.5 Median , $\blacksquare$ because it is not distorted by the outliers of 15,4 k $\blacksquare$ and 44,2 k $\blacksquare \blacksquare$ as it is the case with the mean . 2:O (2)
[24]

6 MATHEMATICAL LITERACY P2 (NOVEMBER 2013)

QUESTION 4

4.1 4.1.1 The dollar is stronger. $\blacksquare$ You have to pay R9,04 for \$1. $\blacksquare$ 1:A
1:R (2)

$$\begin{aligned}
 4.1.2 \text{ Rand value} &= 100 \times 9,04 \blacksquare \\
 &= R904 \blacksquare 1:\text{M}
 \end{aligned}$$

1A (2)

4.1.3 Bank fees are charged when you change from one currency to another . ■■ 2:A (2)

4.1.4 Percentage = $904 - 858,80$ ■
=

x
■

= 5% ■ 1:M

1:M

1:A (3)

4.2 Step 1

With the thumbs push the cover out to remove the battery . ■■

Step 2

Take the battery out . ■■

Step 3

With the battery compartment open and the battery removed, insert the SIM card into the slots with gold plate facing down . ■■

(Accept any relevant explanations .)

2:A

2:A

2:A

(6)
[15]

TOTAL: 100

NATIONAL
SENIOR CERTIFICATE

GRADE 11

NOVEMBER 2013

MATHEMATICAL LITERACY P 2

MARKS: 100

TIME: 2 hours

This question paper consists of 8 pages .

2 MATHEMATICAL LITERACY P2 (NOVEMBER 2013)

INSTRUCTIONS AND INFORMATION

Read the following instructions carefully before answering the questions.

1. This question paper consists of FOUR questions.
2. Answer ALL the questions.
3. Number your answers correctly according to the numbering system used in the question paper.
4. An approved calculator (non-programmable and non -graphical) may be used, unless stated otherwise.
5. ALL calculations and steps must be shown clearly.
6. ALL final answers must be rounded off to TWO decimal places, unless stated otherwise.
7. Write neatly and legibly.

(NOVEMBER 2013) MATHEMATICAL LITERACY P2 3

QUESTION 1

1.1 Harry worked for 3 years (from 2009 –2011) for a company that makes pencil cases. He was retrenched after this period, but he could claim from the UIF, because he contributed to this fund for the years he worked there. He will only be able to claim for a maximum of six months. On retrenchment, Harry 's salary was R 6 240 per month.

UIF = Unemployment Insurance Fund, where an employee pays 1% of his or her earnings together with the employer's 1% every month to the Unemployment Insurance Fund.

1.1.1 Calculate how much money was paid to the UIF per month. (3)

1.1.2 On retrenchment Harry will only be paid a percentage of the salary he earned in the last six months. This percentage is known as the Income Replacement Rate (IRR) and is calculated as follows:

s

What percentage of his monthly salary will be paid to Harry? Give your final answer to 1 decimal place. (3)

1.1.3 Calculate the amount that Harry will receive per month from the UIF. (2)

1.1.4 For how many (full) days will Harry be paid if workers are only entitled to six days they have worked? (3)

1.1.5 Based on your answer in QUESTION 1.1.4, calculate how much Harry will be paid per day. (3)

1.1.6 How much money in total will Harry receive from the UIF for the six month period? (2)

1.1.7 The UIF employee informed Harry that the last payment of the UIF he will receive will be less than the monthly payment that they have calculated. Show by means of calculations that this is TRUE. (3)

4 MATHEMATICAL LITERACY P2 (NOVEMBER 2013)

1.2 The following are pencil cases manufactured by the company Harry worked for. (Not drawn to scale.)

Cylindrical Pencil Case Rectangular Pencil Case
 Height = 178 mm
 Diameter = 95 mm

Surface Area = 2 x Area of Base +
 Perimeter of Base x Height of
 Pencil Case Length = 187 mm
 Breadth = 72 mm
 Height = 38 mm

Surface Area = 2 x Area of Base +
 Perimeter of Base x Height of
 Pencil Case

1.2.1 Calculate the surface area of the cylindrical pencil case in cm². Use the given formula.

Use ■ = 3,142 (4)

1.2.2 Calculate the surface area of the rectangular pencil case in cm². Use the given formula. (4)

1.2.3 A school approached the company to make 600 cylindrical pencil cases. The pencil cases must be of the same colour. A roll of material with dimensions 25 m long and 150 cm wide is given to the workers to make the pencil cases. Show by means of calculation whether the roll will be enough to make all the pencil cases. The following formula will be useful: Area = Length x Breadth (6)
 [33]

(NOVEMBER 2013) MATHEMATICAL LITERACY P2 5

QUESTION 2

2.1 The following is a profile of the Half Marathon of the Two Oceans Marathon race which is held in Cape Town annually. Use the profile to answer the questions below.

2.1.1 What is the distance to be completed for the half marathon? (1)

2.1.2 At what height (altitude) , do the runners start this marathon? (2)

2.1.3 What is the difference between the highest and lowest point on the profile? (2)

2.1.4 With reference to the profile, mention any section that will be the most challenging to the runners. Give a reason for your answer. (3)

2.2 Elena Nugev (a white female's secretary) finished the marathon from 2004 to 2012. The statistics below show the time it took her to finish the marathon from 2004 to 2012.

2004	2005	2006	2007
03:37:51	03:38:12	03:36:29	03:35:34
2008	2009	2010	2011
03:35:25	03:40:43	03:42:19	03:37:54
			03:41:56

2.2.1 In which year did Elena record the best time? Write the time down in

words. (2)

2.2.2 In which year did Elena record the worst time? Write down her worst time. (2)

2.2.3 Describe Elena's finishing time. (2)

2.2.4 Calculate the total time that Elena spent competing in the marathon for the above period. (4)

2.2.5 What is Elena's finishing time? (2)

2.2.6 Calculate the difference between Elena's best and worst finishing times. (3)

6 MATHEMATICAL LITERACY P2 (NOVEMBER 2013)

2.3 Your friend made a statement that in 2013 Elena has a 90% chance of winning the Two Oceans Marathon again.

2.3.1 Explain what is meant with this statement. (2)

2.3.2 Express this probability as a fraction and as a decimal. (3)
[28]

QUESTION 3

3.1 The following is the water and sewage account for Mr Zee. Some of the information is omitted. Study the account and answer the questions below.

Account Details as at 07/01/2013 Account No : 555 444 333
Water (Period 05/12/2012 to 04/01/2013 – 31 days) Actual Reading

At 72 Nowhere Street, Mountain View / ERF 9876543

Previous Reading:

Current Reading:

185 523

Consumption:

30,3 k ■

Consumption Charge

Sewage (Period 05/12/2012 to 04/01/2013 – 31 days) Actual Reading

At 72 Nowhere Street, Mountain View / ERF 9876543

Disposal Charge for k ■: R103,32

VAT at 14%

R

Latest account total due:

R

WATER SEWAGE

Domestic 1: Tariffs Disposal Charges

Volume Tariff per k ■ Volume Tariff per k ■

< 6 k ■ Free < 4 k ■ Free

6,1 – 15 k ■ R7,90 4,1 – 7 k ■ R3,54

15,1 – 25 k ■ R8,90 7,1 – 15 k ■ R8,52

> 25 k ■ R9,90 > 15 k ■ R12,27

3.1.1 Calculate the previous reading. (2)

3.1.2 Use the water tariff table to calculate how much Mr Zee will be charged for his water consumption. (3)

3.1.3 Calculate how many kilolitres were used if the disposal charge is R103,32. (3)

3.1.4 Calculate the VAT amount for this account. (2)

3.1.5 What is the total amount that Mr Zee has to pay to the municipality for this account? (2)

(NOVEMBER 2013) MATHEMATICAL LITERACY P2 7

3.2 A survey regarding the water consumption was conducted in the area where Mr. Zee lives. Twelve households took part in this survey. The following was recorded:

20,8 k■ ; 44,2 k■ ; 22,5 k■ ; 21,6 k■ ; 15,4 k■ ; 22,7 k■ ;
20,4 k■ ; 25,1 k■ ; 23,6 k■ ; 24,9 k■ ; 21,5 k■ ; 30,3 k■

3.2.1 Calculate the mean (average) of water consumption. (3)

3.2.2 Find the median of the data. (3)

3.2.3 Give a reason why you think that the water consumption for one of the households is so low in comparison to the rest. (2)

3.2.4 Give a reason why you think that the water consumption for one of the households is so high in comparison to the rest. (2)

3.2.5 Which of the central tendencies (mean or median) will best describe the data? Give a reason for your answer. (2)

[24]

QUESTION 4

4.1 Zoleka received an unexpected gift in the form of money from a cousin in New York, United States of America. The amount she received is \$100 (US dollars). On the day that Zoleka received the money, she saw that the exchange rate was R 9,04.

4.1.1 Which of the two currencies is the strongest? Give a reason for your answer. (2)

4.1.2 On the day that the exchange rate was R9,04, Zoleka went to the bank to convert her \$ 100. Calculate the value of her dollars in rand. (2)

4.1.3 After Zoleka received her money, she discovered that what she calculated to receive is not the same. She only walked away with R858,80. Why do you think this is the case? (2)

4.1.4 Calculate what percentage of her money she lost. (3)

8 MATHEMATICAL LITERACY P2 (NOVEMBER 2013)

4.2 With the money she changed and other money she saved, Zoleka bought herself a brand new cell phone.

Zoleka wants to insert her SIM card into the phone and have to follow the instructions as shown in the illustrations below.

STEP 1 STEP 2

STEP 3

Briefly explain the steps that Zoleka needs to take to insert the SIM card. (6)
[15]

TOTAL: 100